Physical Chemistry CUME, Nov. 10th, 2017

Quantum

In order to understand the vibrational spectrum of a molecule, one typically approximates a chemical bond as a harmonic oscillator.

(a). Write the time-independent Schrödinger's equation, for a chemical bond with equilibrium bond-length x_e and force constant k. For a diatomic with in single bond, there is only one vibrational degree of freedom, thus for this exercise it is sufficient to write the equation in one-dimension.

(b). The Schrödinger's equation will look more similar to a Harmonic oscillator problem discussed in a standard text books if one replaces x with $\Delta x = x - x_e$, and k with $m \cdot \omega^2$ write the Schrödinger's equation using Δx as the variable. Note you need to first figure out the relationship between $\frac{\partial}{\partial x}$ and $\frac{\partial}{\partial \Delta x}$.

c) Write the energy levels for a Harmonic oscillator using variables, n (quantum number), k (force constant), m (mass or reduced mass)? You do not have to solve the Schrödinger's equation. If you recall this from your memory, writing down the solution directly is fine.

d) The electric field of light at IR frequency can be approximated as uniform on the scale of the molecule. (This is due to the wave-length much larger than the size of a molecule.)

Thus the Hamiltonian for the light-matter interaction can be approximated as $V = \mathrm{E} \cdot \mu$, where E is the external field and μ is the dipole moment. Furthermore it can be shown that the Hamiltonian, V, can be shown to be linear to the sum of the raising and lowering operators a_+ and a_-.

$$V = E \cdot (\mu_0 + C(a_+ + a_-)),$$

where μ_0 is the permanent dipole moment, C is a proportionality constant.

For an eigen-state of the Harmonic oscillator |n>, $a_+ | n >= \sqrt{n+1} | n+1 >$ and $a_- | n >= \sqrt{n} | n-1 >$. (Properties of the raising and lowering operators)

According to time-dependent perturbation theory, the rate of transition from state |n> to state |m> ($m \neq n$) is proportional to the square of $< m | V | n >$, please derive the IR selection rule (whether a transition is allowed) for a harmonic oscillator (an ideal chemical bond) assuming none of the proportionality constants is zero.